

ANTI-RETRO VIRAL DRUGS

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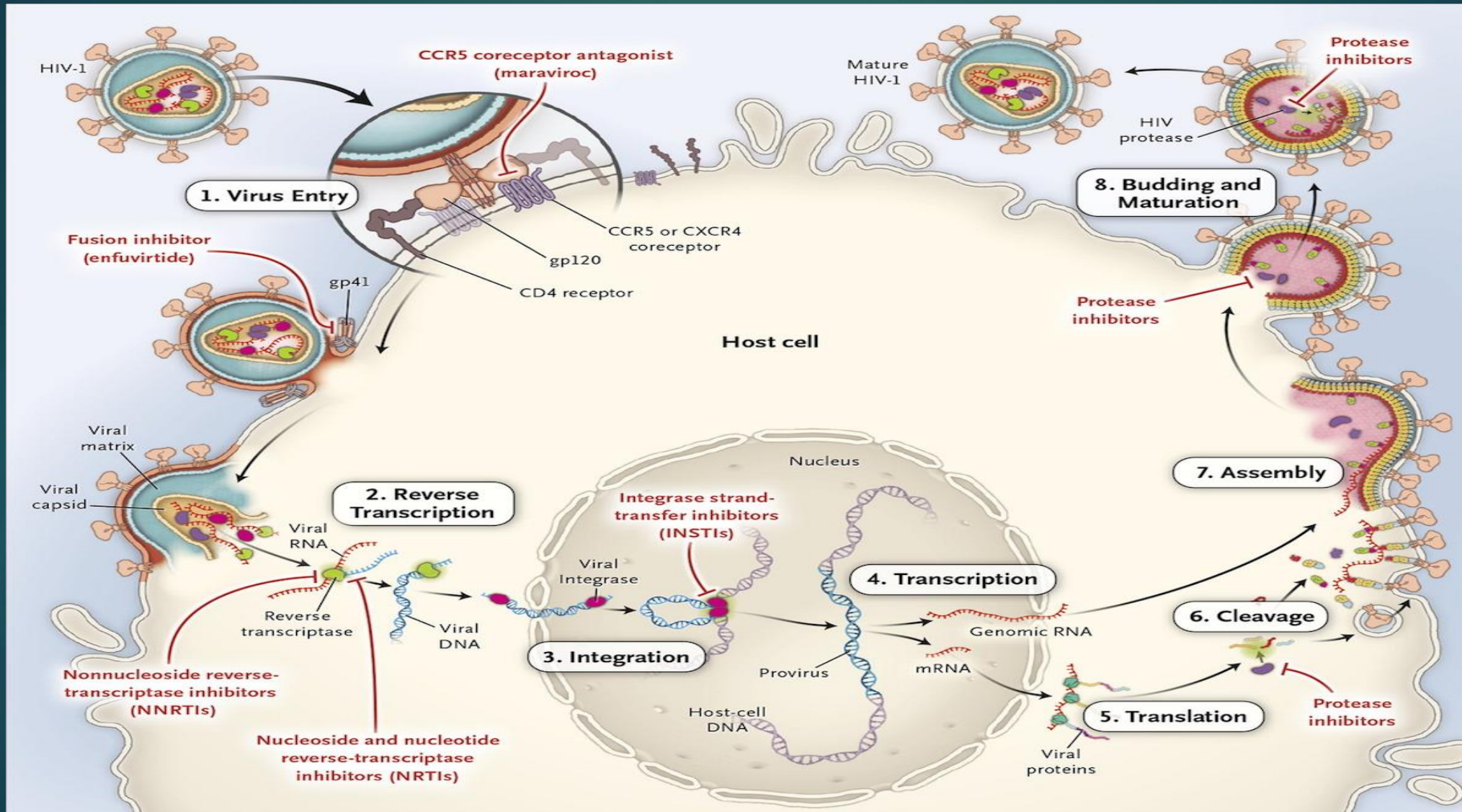
OVERVIEW

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- ▶ HIV is an RNA retrovirus whose replication begins when the **glycoprotein 120** on the surface of HIV 1 binds to CD4 (Cell differentiation) antigen in the surface of the **HIV specific lymphocytes (CD4 cells)**
- ▶ The binding of glycoprotein 120 to CD4 cells causes a conformational change in the glycoprotein 120 enabling it to interact with the **chemokine co-receptor (CCR5 or CXCR4)** on the lymphocyte surface
- ▶ This leads to an exposure of virus fusion protein called **glycoprotein 41** which also undergo conformational changes to permit it insert its hydrophobic tail into the host cell membrane and bind host cell integrins leading to **fusion** of the viral and host cell membranes and transfer of viral genome into the cytoplasm
- ▶ Once HIV enters the CD4 cell, viral RNA serves as a template to produce a complementary doubled stranded DNA in a reaction catalyzed by an enzyme called **reverse transcriptase** (RNA dependent DNA polymerase)
- ▶ The viral DNA then enters the host cell nucleus and is incorporated into the host genome in a reaction catalyzed by an enzyme called **HIV integrase** to produce large non functional polypeptides called **polyproteins**

Life cycle of the HIV and the target sites for antiretroviral drugs

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CLASSIFICATION OF ANTIRETROVIRAL DRUGS

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- ▶ Nucleo(t)side Reverse Transcriptase Inhibitors (NRTIs)
- ▶ Non-nucleoside Reverse Transcriptase Inhibitors (NNRTIs)
- ▶ Protease Inhibitors (PIs)
- ▶ Chemokine (CCR5) co-receptor antagonist (Entry Inhibitors)
- ▶ Fusion Inhibitors
- ▶ Integrase Inhibitors (HIV integrase strand transfer inhibitors)

Nucleo(t)side Reverse Transcriptase Inhibitors (NRTIs)

- ▶ First agents available for HIV Infection
- ▶ Less potent than NNRTIs and PIs
- ▶ Have a central role in ART
- ▶ Have activity against HIV-1 and HIV-2
- ▶ Nucleoside and nucleotide analogues
- ▶ Differ from normal substrates only by a minor modification in sugar (ribose) molecule

Drugs

- ▶ Abacavir (ABC)
- ▶ Didanosine (ddI)
- ▶ Emtricitabine (FTC)
- ▶ Lamivudine (3TC)
- ▶ Stavudine (d4T)
- ▶ Tenofovir (TDF)
- ▶ Zidovudine (ZDV; formerly called azidothymidine [AZT])

Mechanism of action of NRTIs

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- ▶ Reverse transcriptase is also known as an HIV- specific DNA polymerase
- ▶ This enzyme allows HIV RNA to be transcribed into proviral DNA and incorporated into host-cell genome
- ▶ Proviral DNA chain elongation is necessary before genome incorporation can occur
- ▶ Any drug which is able to act as "false" building blocks causes Chain termination and this is what NRTIs do
- ▶ NRTIs once incorporated, work by preventing other nucleosides from also being incorporated
- ▶ Overall, NRTIs interrupt HIV replication cycle via competitive inhibition of HIV reverse transcriptase and termination of the DNA chain

Pharmacokinetics of NRTIs

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- ▶ NRTIs are prodrugs and undergoes phosphorylation by intracellular kinases to exert their activity
- ▶ Oral bioavailability ranges from 25%-93%, with tenofovir and didanosine on lower end of spectrum
- ▶ Food does not significantly affect absorption
- ▶ Except didanosine, which must be taken on empty stomach
- ▶ Renal elimination
- ▶ Exception is abacavir, given at normal dose regardless of creatinine clearance
- ▶ Minimal drug-drug interactions occur
- ▶ Clinically significant Interactions involve didanosine
- ▶ With tenofovir, didanosine levels are higher than expected
- ▶ Didanosine and ribavirin combination should be avoided

Side effects of NRTIs

- ▶ Zidovudine causes bone marrow suppression leading to anaemia
- ▶ Didanosine and stavudine causes pancreatitis and peripheral neuropathy
- ▶ Abacavir is more likely to cause hypersensitivity reactions and hence the need to do a sensitivity test () before treatment
- ▶ Tenofovir causes renal impairment

Non-nucleoside Reverse Transcriptase Inhibitors (NNRTIs)

- ▶ Were introduced in 1996 with approval of nevirapine
- ▶ Have potent activity against HIV-1 and are part of preferred initial regimens
- ▶ Efavirenz, confers most significant inhibition of viral infectivity
- ▶ All exhibit same mechanism of action

Drugs

First-generation

- ▶ Delavirdine (DLV)
- ▶ Efavirenz (EFV)
- ▶ Nevirapine (NVP)

Second-generation

- ▶ Etravirine (ETR)
- ▶ Rilpivirine (RPV)

Mechanism of Action of NNRTIs

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- ▶ These drugs directly inhibit the reverse transcriptase
- ▶ HIV reverse transcriptase is a heterodimer composed of 2 subunits (p66 and p51)
- ▶ NNRTIs bind p66 subunit at a hydrophobic pocket distant from active site of enzyme (allosteric site)
- ▶ This non-competitive binding induces a conformational change in enzyme
- ▶ 1st generation NNRTIs are more rigid in structure
- ▶ Resistance can quickly be developed
- ▶ 2nd generation NNRTIs have a more flexible structure allowing them to adjust readily and resist mutation more effectively
- ▶ Unlike NRTIs, NNRTIs do not require metabolic activation and they are not incorporated into the viral DNA

Pharmacokinetics of NNRTIs

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- ▶ All utilize Cyt P450 for metabolism and exert varying induction and inhibition effects on specific isoenzymes (eg, CYP3A4, CYP2C9)
- ▶ Results in a significant potential for drug-drug interactions
- ▶ Delavirdine primarily uses the 3A4 isoenzyme for metabolism
- ▶ Nevirapine is metabolized mainly by 3A4 with some secondary metabolism through 2B6
- ▶ Efavirenz is primarily metabolized through 2B6 and secondarily through 3A4
- ▶ Etravirine is a substrate of 3A4, 2C9, and 2C19
- ▶ Highly protein-bound (98-99%), primarily to albumin and alpha1 acid glycoprotein except nevirapine
- ▶ Serum half-lives are fairly extended, ranging (25-55 hours) except for delavirdine(2-11 hrs.)

Protease Inhibitors (PIs)

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- ▶ First introduced in 1995
- ▶ Are an integral part of treatment
- ▶ Exhibit activity against clinical isolates of both HIV-1 and HIV-2

Drugs

- ▶ Atazanavir sulfate, ATV
- ▶ Darunavir
- ▶ Fosamprenavir Calcium, FOSAPVIndinavir, IDV
- ▶ Lopinavir / ritonavir, LPV/RTV
- ▶ Nelfinavir mesylate, NFV
- ▶ Saquinavir mesylate, SQV
- ▶ Tipranavir, TPV

- PIs are synergistic to NRTIs and are often combined in treatment regimen
- Because ritonavir inhibits the metabolism of other PIs, it is usually combined with other PIs to increase their plasma levels and duration and this is referred to as **boosted therapy**

Pharmacokinetics

- ▶ Significant first-pass metabolism by cytochrome P450 (CYP) 3A4 and 3A5 and intestinal efflux by p-glycoprotein is observed
- ▶ Highly protein-bound (97-99%), primarily to albumin and alpha1 acid glycoprotein except indinavir
- ▶ Short serum half-lives, ranging from 1.5 - 2 hours for indinavir and 7 hours for atazanavir
- ▶ Significant Interactions with medications cleared through CYP450 isoenzymes
- ▶ Low-dose ritonavir (100-200 mg) is frequently co-administered with other protease inhibitors to block intestinal and hepatic 3A metabolism

Entry Inhibitors (Chemokine CCR5 co-receptor antagonist)

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- ▶ Example of is **Maraviroc**
- ▶ The drug has been found to be active against HIV strains found to be resistant to NRTIs and PIs and has shown to have synergistic effects with enfuvirtide, a fusion inhibitor
- ▶ These inhibit the entry of HIV
- ▶ Binding of gp120 HIV surface protein to CD4 receptor induces a structural change that reveals V3 loop of the protein
- ▶ V3 loop then binds with a chemokine co-receptor (principally either CCR5 or CXCR4), allowing gp41 to insert itself into the host cell and leading to fusion of the cell membranes
- ▶ Maraviroc selectively and reversibly binds CCR5 co-receptor, blocking V3 loop interaction and inhibiting fusion of cellular membranes
 - As some viral strains may use an alternate co-receptor CXCR4 for entry
 - A tropism assay is necessary to confirm that patient's virus only uses CCR5 for entry

Fusion inhibitors

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- ▶ An example is **Enfuvirtide**
- ▶ Act extracellularly to prevent fusion of HIV to CD4 or other target cell
- ▶ Blocks second step in fusion pathway by binding to HR1 region of gp41
- ▶ Does not allow HR1 and HR2 to fold properly
- ▶ Thus preventing conformational change of gp41 required to complete final step in fusion process
- ▶ Dose adjustments are not required in patients with renal insufficiency or mild-to-moderate hepatic insufficiency

Adverse drug reactions

- ▶ Injection-site reactions (e.g. pain, erythema, induration, nodules) diarrhea, nausea, fatigue, hypersensitivity reactions, increased rate of bacterial pneumonia

Integrase Inhibitors (HIV integrase strand transfer inhibitors)

- ▶ Examples are Raltegravir & Elvitegravir , Dolutegravir
- ▶ HIV integrase is responsible for transport and attachment of proviral DNA to host-cell chromosomes, allowing transcription of viral proteins and subsequent assembly of virus particles
- ▶ Integrase inhibitors competitively inhibit strand transfer reaction by binding metallic ions in active site

Pharmacokinetics of integrase inhibitors

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Raltegravir

- ▶ Rapid absorption, taken with or without food, half-life of 10-12 hours
- ▶ Longer half-life in women
- ▶ 83% bound to plasma proteins
- ▶ Other antiretroviral agents may alter metabolism
- ▶ Antacids may decrease absorption by divalent cation binding

Elvitegravir

- ▶ Administered with low-dose ritonavir (100 mg) to reduce its first-pass metabolism and systemic clearance
- ▶ Co-administration results in a 20-fold increase in systemic exposure and a terminal half-life of 10-13 hours
- ▶ Metabolized through CYP3A4 and UGT1A1/UGT1A3
- ▶ Drug-drug interactions with other medications are likely because of ritonavir
- ▶ Antacids may decrease absorption

Drugs with Potential to Interact with PIs or NNRTIs

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- ▶ Statins (simvastatin & lovastatin)
- ▶ Azole antifungals
- ▶ Anticonvulsants
- ▶ Anti-TB (Rifampicin)
- ▶ Warfarin
- ▶ Midazolam
- ▶ Trizolam
- ▶ Clarithromycin
- ▶ Oral contraceptives
- ▶ Amitriptyline

GOALS OF ANTI RETROVIRAL THERAPY

- ▶ **Increasing of the CD4 cells** - Prevention or delay of progressive immunodeficiency
- ▶ **Reduction of viral load** - Control of viral replication
- ▶ Prevention of opportunistic infection - Delayed progression to AIDS and thus leading to prolonged survival
- ▶ Eventual Improved quality of life of a patient

Consideration for treatment

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Earlier guidelines

Previously the decision to initiate HIV treatment was based on several considerations as below

- ▶ CD4 count of less than 500 CD4 cells/microliter
- ▶ Viral load of 100,000 and above copies per milliliter regardless of the CD4 count
- ▶ Patients with an AIDS defining illness

Current guidelines

- ▶ Currently the guidelines recommend the “**Test**” and “**Treat**”
- ▶ This current model recommends that all those patients who test HIV positive should be commenced on treatment regardless of their CD4 count and viral load
- ▶ Improved therapy outcome have been noted in the test and treat model

DRUG REGIMENS FOR TREATING HIV

- ▶ The preferred regimen for commencing HIV treatment consist of two NRTIs and either an NNRTI or PI
- ▶ The NNRTI based regimen is the simplest for patients
- ▶ PI based regimen usually require a ritonavir boosted dose
- ▶ To determine the response, viral loads are measured 3 to 6 months after initiation of therapy

Change of treatment regimen

- ▶ Reason for changing treatment regimens are either treatment failure or toxicity

Types of Treatment Failure

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1. Virologic Failure:

- ▶ if viral load is not <400 copies/mL after 3 to 6 months

2. Immunologic Failure

- ▶ The CD4 cell count persistently falls below the baseline CD4 cell count
- ▶ The CD4 cell count fails to increase by more than 25-50 cells/ μ L after one year of treatment
- ▶ There is a $> 50\%$ decline in CD4 cell count from its highest level on ART

3. Clinical Failure

- ▶ when the patient has a new AIDS-defining illness mostly conditions graded as stage 3 or 4 as defined by WHO after initiation of ART e.g. Cryptococcosis, toxoplasmosis, CMV, Kaposi sarcoma

Serious Adverse effects of NRTIs

- ▶ Lactic acidosis/fatty liver
- ▶ Lipoatrophy (loss of subcutaneous fat)
- ▶ Anemia
 - Zidovudine (AZT)
- ▶ Pancreatitis
 - Didanosine (ddl)
- ▶ Neuropathy
 - Didanosine (ddl)
 - Stavudine (d4T)

Serious Adverse Effects of NNRTIs

- ▶ All NNRTIs can cause the hepatitis and skin rash
 - Hepatitis
 - Skin rash
- ▶ CNS symptoms
 - Efavirenz
- ▶ Stevens-Johnson syndrome*
 - Nevirapine

Serious adverse Effects of PIs

- ▶ All PIs can cause the following
 - Insulin resistance hyperglycemia and diabetes
 - Elevated serum lipids
 - Abnormal fat accumulation
 - Liver toxicity

LIPODYSTROPHY SYNDROME

- ▶ Main clinical features are peripheral fat loss, central
- ▶ Fat accumulation, gynaecomastia, buffalo hump and other peripheral lipomatosis
- ▶ Incidence: 20-80% of pts in ARV drugs with PIs being the leading cause of this adverse
- ▶ Presumed Mechanism: inhibition of DNA polymerase gamma resulting in depletion of mitochondrial DNA

Laboratory monitoring of patients on ART

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CD4 cell count %:

- ▶ At 3 and 6 months post-initiation, then every 6 months (all ages)

Viral load:

- ▶ At 3 and 6 months post-initiation, then as follows: (Every 6 months for adults)

FBC:

- ▶ AZT-based ART: at 4 and 12 weeks post-initiation, then annually only, and as clinically indicated • If not on AZT-based ART: annually only, and as clinically indicated Laboratory

Laboratory monitoring of patients on ART Cont'd

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AST/ALT:

- ▶ NVP-based ART: 2, 4, and 12 weeks post-initiation, thereafter only as clinically indicated
- ▶ EFV-based ART: at 4 and 12 weeks post-initiation, thereafter only as clinically indicated
- ▶ PI-based ART: only as clinically indicated

Glucose and total cholesterol/triglycerides

- ▶ Annually only if on PI based ART

Creatinine and creatinine clearance :

- ▶ At 3 and 6 months post-initiation and then, if stable, every 6 months (TDF only)

RPR (rapid plasma reagen) or VDRL test:

- ▶ After baseline, only as indicated

Opportunistic infections (OI's)

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- ▶ These are infections which do not usually occur in immunocompetent individuals but are common in immune compromised patients
- ▶ Occurrence of such infections help in grading the level of HIV based on WHO staging as different OI's usually occur at different levels of compromised state (CD4 count)
- ▶ Below are examples of OI's;
- ▶ Vaginal candidiasis
- ▶ Skin diseases
- ▶ Fatigue
- ▶ Bacterial pneumonia
- ▶ Herpes zoster
- ▶ Kaposi sarcoma
- ▶ Pneumocystis jiroveci pneumonia (PJP)
- ▶ Toxoplasmosis
- ▶ Cryptocosis
- ▶ CMV
- ▶ MAC
- ▶ CNS lymphomas

Principles of HIV Drug Resistance

- ▶ Results from changes (mutations) in genetic information in virus
- ▶ These changes occur whenever HIV is replicating
- ▶ Partial HIV suppression promotes resistance
- ▶ Resistance can be delayed by suppressing virus completely
- ▶ Reverse transcriptase and protease are flexible (highly mutable)
- ▶ Resistance may fade but not disappear when a drug is stopped
- ▶ Some mutations allow certain viruses to resist effects of one or more antiretroviral drugs
- ▶ Drug resistant virus usually grows faster and better than drug susceptible virus
- ▶ Drug resistant virus replaces drug susceptible virus in patients

HIV PROPHYLAXIS

- ▶ This administration specific regimen of antiretroviral drugs to prevent transmission of HIV and further prevent the subsequent HIV infection
- ▶ The different types of prophylaxis under HIV treatment are;
 1. Prevention of Mother to Child Transmission (PMTCT)
 2. Pre - Exposure prophylaxis (PREP)
 3. Post Exposure prophylaxis (PEP)

PREP

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- ▶ PrEP is a daily course of ARVs taken by HIV-negative people who may be at substantial risk of HIV acquisition as part of combination HIV prevention approaches
- ▶ PrEP involves the use of antiretroviral (ARV) drugs before HIV exposure by people who are not infected with HIV to block the acquisition of HIV
- ▶ To gauge the effectiveness of PREP, many trials were conducted among the known risk groups like sero-discordant couples, heterosexual men, women, men who have sex with men, people who inject drugs, transgender men and women
- ▶ Significant levels of efficacy was achieved where adherence was very high adherence

PrEP Cont'd

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- ▶ Among the known regimens for PrEP are;
 - Oral drugs containing Tenofovir Disoproxil Fumarate (TDF) with either Emtricitabine (FTC) or Lamivudine (3TC)
 - Alternatively, TDF is replaced with Tenofovir alafenamide (TAF) in patients with renal insufficiency
 - Recently Carbotegravir, an Integrase inhibitor has been introduced as an injectable form which offers protection for 2 months
- PrEP clients must be routinely tested for HIV infection, and ART offered immediately if the PrEP user seroconverts (Tests HIV positive)
- PrEP alone is not 100% effective at preventing HIV and clients need to be counselled that it should be used in combination with other HIV prevention methods as well

PEP

- ▶ Post-Exposure Prophylaxis (PEP) is the use of ART to prevent HIV transmission after occupational exposure to HIV-infected blood
- ▶ Non-occupational exposure to HIV in children and adolescents is mostly attributed to sexual abuse
- ▶ PEP should be commenced in a period not exceeding 72 hours after exposure

- ▶ Globally mother to child transmission of HIV is a significant contributor (9%) to paediatric HIV infections. In Africa more than 95% of paediatric HIV infections is due to mother to child transmission of HIV
- ▶ Preventive measures are started before pregnancy and then scheduled reviews are done and the have suitable responsive steps taken
- ▶ An HIV positive woman who is pregnant is followed up and guided on how to carry themselves in terms of their sexual life even as they are put on treatment (DTG + TDF/TAF + XTC)

PMTCT Cont'd

- ▶ The desire is to have infants be born HIV negative but in case they are born positive put on appropriate regimen like we have those who may be positive e.g. AZT+3TC+NVP - until 4 weeks old and weight ≥ 3 kgs and thereafter change to the ABC+3TC+DTG
- ▶ Those born negative are put on prophylaxis and followed up at 6 weeks, 10 weeks, 14 weeks, 6 months, 9 months, 12 months, 18 months and 24 months
- ▶ The prophylactic regiment is AZT+3TC+NVP

Regimens for PEP

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- ▶ Preferred PEP regimen : TDF or TAF + XTC + DTG

Alternative

- ▶ TDF or TAF + XTC + DRV-r
- ▶ TDF or TAF + XTC + LPV-r TDF or TAF + XTC + ATV-r
- ▶ AZT + 3TC + LPV-r (children < 20kg)
- ▶ AZT + 3TC + DTG (children ≥ 3kg Where available)
- ▶ TAF + FTC + DTG (children ≥ 25kg)

NOTE:

- XTC represents either lamivudine or emitricitabine
- Clients on PEP should have an HIV test before starting PEP, 6 weeks and at 3 months

Resistance Testing

There are two types of resistance :

i. Genotyping

- ▶ Detects drug resistance mutations on virus genome that may make it resistant to certain anti-retrovirals
- ▶ Less expensive
- ▶ Can usually be completed in 1-2 weeks

ii. Phenotyping

- ▶ Measure ability of viruses to grow in presence of various concentrations of antiretroviral drugs
- ▶ More expensive
- ▶ Generally takes 2-3 weeks to complete

Resistance Mutations

- ▶ For some drugs (NNRTIs and 3TC), a single mutation causes high-level resistance
 - ▶ – Resistance to these drugs occurs very quickly
- ▶ • For other drugs (most NRTIs and PIs), many mutations must occur before high-level resistance is observed.
 - ▶ – Resistance to these drugs occurs more slowly

Cross-Resistance

- ▶ Resistance to one drug can cause resistance to others of the same class
 - NNRTI: complete cross-class resistance
 - NRTI: partial cross-class resistance
 - PI: partial cross-class resistance
 - Combination with low ritonavir boosting dose partly overcome this

Minimization of emergence of Viral Resistance

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- ▶ ARVs should not be prescribed in absence of adherence counselling and support
- ▶ ARVS should never be prescribed as monotherapy or dual therapy but should always be triple therapy
- ▶ Serum drug concentrations should be optimized and hence appropriate doses should be given
 - Appropriate doses should always be prescribed and administered
 - Interactions should be avoided by all means possible
 - Mal absorption should be diagnosed and managed
- ▶ If ARV medications are to be discontinued, they should all be stopped at the same time with the possible exception NNRTI based regimen

END